

Lecture-19

HT for Means, Variance & Proportions

- reject H_0 if $p \leq \alpha$
- 1 populaⁿ mean, when variance is known
 $X_1, \dots, X_n, \mu, \sigma^2$

$$H_0: \mu = \mu_0$$

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

$$Z \sim N(0, 1)$$

$$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$$

$$\bar{X} \sim N(\mu_0, \frac{\sigma^2}{n})$$

- direcⁿ of the rejectⁿ req. comes from H_1 .

Right-tail: $\rightarrow H_0: \mu \leq \mu_0 \quad H_1: \mu > \mu_0$

Reject H_0 if

$$Z > Z_\alpha$$

$$P(Z > Z_\alpha) = \alpha$$

eg. $Z_{0.05} = 1.645$

$Z > 1.645 \Rightarrow \text{reject } H_0$

left-tail: $\rightarrow H_0: \mu = \mu_0 \quad H_1: \mu < \mu_0$

Reject H_0 if

$$Z < -Z_\alpha$$

two-sided: $\rightarrow H_0: \mu = \mu_0 \quad H_1: \mu \neq \mu_0$

Reject H_0 if $|Z| > Z_{\alpha/2}$

eg.

$n = 100, \bar{X} = 71.8, \sigma = 8.9, \alpha = 0.05$

$H_0: \mu = 70, \quad H_1: \mu > 70 \text{ (right)}$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{1.8}{8.9/\sqrt{100}} = \frac{1.8}{0.89} = 2.02$$

$\alpha = 0.05, \quad Z_\alpha = 1.645$

$Z(2.02) > Z_\alpha(1.645)$

$\therefore$, reject H_0 .

or

$p = (P(Z \geq 2.02)) = 0.0217$

$p \leq \alpha$

$0.0217 \leq 0.05 \checkmark$

reject H_0

- Right $H_0: \mu = \mu_0$, $H_1: \mu > \mu_0$
 $p\text{-value} \Rightarrow \boxed{P(Z \geq Z_{\text{obs}})}$

left $H_0: \mu = \mu_0$, $H_1: \mu < \mu_0$
 $p\text{-value} \Rightarrow \boxed{P(Z \leq Z_{\text{obs}})}$

two-side $H_0: \mu = \mu_0$, $H_1: \mu \neq \mu_0$
 $p\text{-value} \Rightarrow \boxed{2P(Z \geq |Z_{\text{obs}}|)}$

Eg. $\mu_0 = 8$, $\sigma = 0.5$, $n = 50$, $\bar{x} = 7.8$, $\alpha = 0.01$
 $H_0: \mu = 8$, $H_1: \mu \neq 8$ (two-sided)
 $Z_{0.005} = 2.576$

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{7.8 - 8}{0.5/\sqrt{50}} = \frac{-0.2}{0.5/\sqrt{50}} = \frac{-0.2}{0.07071} = -2.828$$

$$|Z| > Z_{\alpha/2} \Rightarrow |-2.828| > 2.576 \quad \checkmark$$

reject H_0

or $p = 2P(Z \geq |Z_{\text{obs}}|) = 0.0046 \leq \alpha$
0.01 $\checkmark$

- 2-sided HT & CI gives the same decision.

$H_0: \mu = \mu_0$ & $H_1: \mu \neq \mu_0$
 $\boxed{\text{reject } H_0 \Leftrightarrow \mu_0 \notin (1-\alpha) \text{ CI}}$

$$\text{CI} = \bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$\text{HT} \Rightarrow \left| \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \right| \leq Z_{\alpha/2} \Rightarrow |\bar{x} - \mu_0| \leq Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$\bar{x} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu_0 \leq \bar{x} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- if σ is unknown $\Rightarrow Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \rightarrow \boxed{T = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}}$

$T \sim t_{n-1}$ $\boxed{\nu = n-1}$

$\therefore$ t distribution has heavier tails than normal distribution

- $X_1, \dots, X_n$ & $Y_1, \dots, Y_n$ $H_0: \mu_1 = \mu_2$
if σ_1^2, σ_2^2 are known, $H_0: \mu_1 - \mu_2 = d_0$

$\bar{X}_1 - \bar{X}_2$ has var. $\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$H_1: \mu_1 - \mu_2 > d_0$$

$$\Rightarrow Z > Z_{\alpha}$$

- 2 means, unknown but equal var.

$$\sigma_1^2 = \sigma_2^2 = \sigma^2$$

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$v = n_1 + n_2 - 2$$

- 2 means, unknown & unequal var.

$$T' = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

$$v = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\left(\frac{S_1^2/n_1}{n_1 - 1}\right)^2 + \left(\frac{S_2^2/n_2}{n_2 - 1}\right)^2}$$

- paired obs. (not independent)

$$D_i = X_i - Y_i$$

$$\bar{D} = \frac{1}{n} \sum D_i, S_D$$

$$H_0: \mu_D = d_0$$

$$T = \frac{\bar{D} - d_0}{S_D / \sqrt{n}}$$

$$v = n - 1$$

- 1 populaⁿ mean $\Rightarrow \sigma$ known $\rightarrow Z$ test

σ unknown $\rightarrow t$ test

2 indep. means $\Rightarrow$ known var. $\rightarrow Z$

$\Rightarrow$ unknown but = var. $\rightarrow$ pooled t

$\Rightarrow$ " & $\neq$ var. $\rightarrow$ welch t

matched pairs $\rightarrow$ paired t

- $X_i = \begin{cases} 1 & \text{success} \\ 0 & \end{cases}$ $X = \sum_{i=1}^n X_i$ (no. of success)
 $X \sim \text{Bin}(n, p)$
 $H_0: p = p_0$ $H_1: p > p_0$ or $p < p_0$ or $p \neq p_0$

- small sample proportion test

$$X \sim \text{Bin}(n, p_0)$$

right $\Rightarrow H_1: p > p_0 \Rightarrow \boxed{p\text{-value} = P(X \geq x | p = p_0)}$

left $\Rightarrow H_1: p < p_0 \Rightarrow \boxed{p\text{-value} = P(X \leq x | p = p_0)}$

two $\Rightarrow H_1: p \neq p_0 \Rightarrow$ look at both tails

- for large n , $H_0: p = p_0$

$$E(X) = np_0$$

$$\text{Var}(X) = np_0(1-p_0)$$

$$Z = \frac{X - np_0}{\sqrt{np_0(1-p_0)}}$$

$$\hat{p} = \frac{X}{n}$$

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

$\rightarrow$ if n is large binomial becomes normal

- $\hat{p}_1 = \frac{x_1}{n_1}$, $\hat{p}_2 = \frac{x_2}{n_2}$

$$H_0: p_1 = p_2$$

$$SE(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

total success $\Rightarrow x_1 + x_2$

total obs. $\Rightarrow n_1 + n_2$

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

- 1 population var.

$$X_1, \dots, X_n \sim N(\mu, \sigma^2)$$

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$

$$H_0: \sigma^2 = \sigma_0^2$$

for $H_0 \Rightarrow \boxed{\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}}$

$\rightarrow \chi^2_{n-1}$ $\nu = n-1$

chi-sq $\rightarrow$ non-negative, right-skewed

- right: $H_0: \sigma^2 = \sigma_0^2$, $H_1: \sigma^2 > \sigma_0^2 \rightarrow \text{reject}$
left: $H_1: \sigma^2 < \sigma_0^2 \rightarrow \text{reject}$
two-sided: $H_1: \sigma^2 \neq \sigma_0^2$

- 2 populaⁿ var.

$$\sigma_1^2, \sigma_2^2 \quad H_0: \sigma_1^2 = \sigma_2^2 \quad F = \frac{\sigma_1^2}{\sigma_2^2}$$

is + 2 right $\leftarrow F \sim F_{n_1-1, n_2-1}$
skewed

- $Z \rightarrow$ means with known var., large
- sample properⁿ

$t \rightarrow$ means with unknown var.

$\chi^2 \rightarrow$ 1 var.

$F \rightarrow$ ratio of 2 variances

binomial $\rightarrow$ small sample, exact properⁿ
test

- populaⁿ $\rightarrow \mu, \sigma^2, p$ & statistics $\rightarrow \bar{X}, s^2, \hat{p}$
never say $H_0: \hat{p} = 0.5$ always
 $H_0: p = 0.5$

- large n helps do rejections easier.

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

as $n \uparrow$, $\sqrt{n} \uparrow$, $\sigma/\sqrt{n} \downarrow$

- steps

1. identify $\rightarrow \mu, \mu_1 - \mu_2; p, p_1 - p_2, \sigma^2, \sigma_1^2/\sigma_2^2$

2. write H_0 & H_1

3. identify tail ($>$) right, ($<$) left, ($\neq$) (two)

4. choose the test statistic

5. calc. test statistic

6. compute p -value see critical reg.

7. $p \leq \alpha \rightarrow \text{reject } H_0$